

Shihab Bin Kader

Computer Science and Engineering Graduate

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RESEARCH INTEREST

Computer Vision, Deep Learning, and Explainable AI (XAI). Particularly interested in developing interpretable hybrid deep learning models for real-world applications in precision agriculture and smart farming systems.

EDUCATION

East Delta University,

Bachelor of Science in Computer Science and Engineering

20/09/2020 – 18/01/2025

Chittagong, Bangladesh

- Number of credits: 162
- Final grade: 3.13/4.00

• **Thesis: Mango Ripeness Detection Using Deep Learning with XAI (GradCAM)**

Relevant Coursework

- Programming Languages (C, C++, Python, Java)
- Data Structures and Algorithms
- Database Management Systems
- Artificial Intelligence and Machine Learning
- Computer Networks
- Software Engineering
- Image Processing

HONOURS AND AWARDS

Dean's List Awards, *East Delta University*

- Recognized for outstanding academic achievement by being consecutively named to the Dean's List at the institution

EDU Academic Scholarship, *East Delta University*

- Awarded a tuition grant by EDU for exceptional performance during undergraduate studies

PUBLICATIONS

Hybrid Xception-LSTM Model for Explainable Mango

11/2025

Ripeness Classification Using Grad-CAM, 2025 IEEE 4th

International Conference on Robotics, Automation,

Artificial-Intelligence and Internet-of-Things

Developed a hybrid Xception-LSTM model for mango ripeness classification achieving 98% accuracy on a 6,000-image dataset with 6 ripeness stages. Enhanced model interpretability using Grad-CAM visualizations.

SKILLS

Programming Languages (Python, Java, C, C++) | **Deep Learning** (Xception, LSTM, Hybrid CNN-RNN Architectures, Transfer Learning) | **Tools** (TensorFlow/Keras, GitHub, MySQL, Android Studio, HTML & CSS)

PROJECTS

Sign Sense, *Web application for sign language learning*

- Built a Python web application with interactive tutorials for sign language alphabet and basic phrases
- Implemented user authentication and personalized progress tracking system
- Deployed the application on cloud platform for scalability

Fun with Flags, *Mobile application quiz*

- Led development of an Android flag quiz app using Java and Android Studio
- Designed and implemented backend using MySQL for efficient data management
- Added score tracking, result reporting, and user feedback features

LANGUAGE SKILLS

Bengali (Mother tongue) | **English**

VOLUNTEERING

Community Volunteer

Present | Chittagong

- Volunteered in community service activities supporting children through educational and social programs, helping organise events and contributing to initiatives that promote child welfare and development.